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Review Paper:

Climate Change and its diverse impact on The Rural Infrastructures in Bangladesh

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Abstract

Bangladesh is a disaster-prone country which is vulnerable to climatic risks. Climate change has a great impact on rural physical infrastructures. The physical infrastructures include roads, bridges/culverts, markets, disaster shelters, solid waste disposal and small-scale water resources. Due to climate change, the physical infrastructure will suffer from damage after natural disasters, as climate change amplifies the chance of natural disaster. The salinity problem, soil erosion, embankment failure due to soil conditions and disaster issues greatly affect the rural infrastructure. Furthermore, due to the impacts, the repair, operations and maintenance costs of rural infrastructure will be increased. As higher design standards will be required to build climate proof infrastructures, costs of new infrastructure projects will be increased.

This study will be helpful for policy makers, infrastructure planners, stakeholders and implementing agencies like the Local Government Engineering Department (LGED) to deploy disaster resilience infrastructures by considering the effect of climate change on rural infrastructures.

Keywords: Climate Change, Rural Infrastructure, Design standard, Vulnerable.

Introduction

Almost every year, Bangladesh experiences disasters of one kind or another such as tropical cyclones, storm surges, coastal erosion, floods, and droughts causing heavy loss of life and property and jeopardizing the development activities. The country is already beset with many problems like high population density (164.7 million people living in an area of 147570 km²), shortage of land to accommodate the people, food security, human health, illiteracy, and so forth. The above-mentioned types of disasters make the problems all the more complicated.

In the foreseeable future, Bangladesh is likely to be one of the most vulnerable countries of the world in the event of climate change. Global warming due to the increase in greenhouse gas concentrations in the earth's atmosphere and the consequent sea level rise (SLR) are going to add fuel to

the fire. Almost every sector of socio-economic life in Bangladesh is likely to be affected by climate change.

Most of the climate change impacts in Bangladesh are likely to come from the south i.e. the Bay of Bengal and the adjoining North Indian Ocean. These waters are the sources of climate-induced hazards such as tropical cyclones and storm surges, coastal erosion, monsoon wind, evaporation for monsoon rainfall, floods, and droughts, frequently impact families and livelihoods while also damaging critical rural infrastructure including water supply and drainage, embankments, roads and bridges. These damages leave rural populations without basic services and often in full isolation.

LGED should focus on mainstreaming climate change in all its infrastructure design, implementation and maintenance programme so that changing climate does not jeopardize our continued economic growth. Surprisingly, literature review shows that relevant studies are very limited especially in the context of developing countries. Targeting Bangladesh, one of the most vulnerable countries in the world to climate and the sixth most vulnerable to floods, this study shows how climate change may damage rural infrastructures, to the potential detriment of economic growth, particularly in developing countries like Bangladesh.

Vulnerability mapping, climate monitoring, prediction and timely early warning of the extreme climate events are therefore some of the best strategies for mitigating their negative impact on humanity, property and the environment as well as taking advantage of any positive impact. Accurate and timely information on the characteristics of the extreme climate events including the associated vulnerabilities is therefore crucial input in sustainable development planning. Moreover, road infrastructure not only mitigates poverty, but also is the valid pathway of this area of intervention to poverty reduction at the household-level.

Review of Literature

The Intergovernmental Panel on Climate Change (IPCC) has already noted that Bangladesh will be among the worst victims due to climate change. A rise in both sea level and frequency of cyclone-storms due to the escalating atmospheric temperature is likely to have grave food and energy security consequences, which will also facilitate different types of diseases and frequency of other natural calamities. Weather changes in extreme and climate conditions have important impacts that square measure the foremost serious challenges to society¹. Temperature and

precipitation variability are the very important sides influencing climate variability and extremes².

According to the IPCC (2007), due to continuing warming as a result of the climate changes, the world population can become rather more sufferer. Bangladesh, a south-Asian domain, is highly ranked within the list of most climate-vulnerable countries on earth^{3,4}. High temperatures, heavy rainfall and seasonal variation square measure the distinctive characteristics that distinguish the climate of Bangladesh from that of other tropical regions⁵. Different climate changes occur in Bangladesh such as river bank erosion, floods, downing ground water level, increasing salinity, drought in the rainy season which have been contributing to increase the vulnerability of many regions.

The information on long term climatic trends is infrequent and insufficient in Bangladesh⁶. The rate of temperature increase will be higher than the present⁷. Throughout the past 100 years, a rise of temperature over Bangladesh is 0.5°C⁸. Another study shows that the mean annual temperature of Bangladesh has increased during the period of 1895-1980 at 0.3°C over the past two decades^{9,10}. In general, associated degrees of increasing trend were found in both summer and winter temperatures¹¹. Annual mean maximum temperature will increase to 0.4°C and 0.73°C by the year of 2050 and 2100 severally¹².

Climate change indeed is real. Super typhoon Haiyan is the latest natural disaster that has also led credence to the reality of climate change. This sad occurrence hit land and devastated the Philippines. This record-breaking storm is the strongest storm in history to make landfall. It tore apart buildings and left entire provinces without power or communication. Walls of water as high as fifteen feet swept over the country washing away towns on many islands and washed ships ashore where homes once stood²⁹.

Climate change is a serious risk to poverty reduction and could undo decades of development efforts. While climate change is global, its negative impacts are more severely felt by poor people and poor countries. They are more vulnerable because of their high dependence on natural resources and limited capacity to cope with climate variability and extremes.

Country Background: Bangladesh is located between 26° North and 88° to 92° East. It is bordered on the west, north and east by India, on the south-east by Myanmar, and on the south by the Bay of Bengal. Most of the country is low-lying land comprising mainly the delta of the Ganges and Brahmaputra rivers. Floodplains occupy 80% of the country. Mean elevations range from less than 1 meter on tidal floodplains, 1 to 3 meters on the main river and estuarine floodplains, and up to 6 meters in the Sylhet basin in the north-east³⁰. Only in the extreme northwest elevations are greater than 30 meters above the mean sea level. The

northeast and southeast portions of the country are hilly with some tertiary hills over 1000 meters above mean sea level³¹.

Bangladesh ranks low on just all measures of economic development. This low level of development, combined with other factors such as its geography and climate, makes the country quite vulnerable to climate change. Bangladesh is a very densely populated country with 75% of the population living in rural areas³². About one-quarter of the country's GDP comes from agriculture³² which makes the country's economy relatively sensitive to climate variability and change.

It is difficult to determine Bangladesh's potential to adapt to climate change, but several key statistics give some insight as to the state of its infrastructure and social and human capital. In 2000, the World Bank estimated that only 9.5% of Bangladesh's 207,500 km network of roads was paved, putting it well below the average for low income countries of 16.5%³², suggesting that its physical infrastructure in general might be less developed than that of low-income countries. In the same year, the World Bank reported Bangladesh had only 51 scientists and engineers per million people, a number comparable to that for low income countries in general. Similarly, gross secondary and tertiary school enrollment stood at 47.5% and 4.8% respectively in 2000. A relatively uneducated and illiterate public will be less capable of adapting to climate change, and thus has higher vulnerability.

Consequences of Global Climate Change: Impact Of global climate change is already visible, for instance, temperatures are rising, polar caps are melting, sea level is rising, the desertification increases and the winters in Europe become ever wetter. It is also concluded that the number of natural disasters increases more and more. Tsunamis, floods and extreme drought occur more frequently than in times past. In the period 1950-1960 worldwide, 13 natural disasters have been registered against 72 in the period 1990-1998. Now already the consequences are clearly demonstrable and most likely they will only increase in extent and frequency in the future.

In the course of the present century, sea levels will rise by between 18 and 59 centimeters. This is because warmer water occupies more space than cold water and because of the retreat of glaciers and polar ice sheets. Our understanding of the melting of the Greenland and Antarctic ice sheets is still incomplete. This, together with the fact that there may be large regional variations in sea level rise, means that in some parts of the world, the consequences may be even more dramatic than predicted by IPCC.

Climate Change in Bangladesh- Key Impacts and Vulnerabilities

Bangladesh has a humid, warm, tropical climate. Its climate is influenced primarily by monsoon and partly by pre-monsoon and post-monsoon circulations. The south-west

monsoon originates over the Indian Ocean and carries warm, moist and unstable air. Climate change poses significant risks for Bangladesh, yet the core elements of its vulnerability are primarily contextual. Between 30-70% of the country is normally flooded each year. The huge sediment loads brought by three Himalayan rivers, coupled with a negligible flow gradient add to drainage congestion problems and exacerbate the extent of flooding.

The societal exposure to such risks is further enhanced by Bangladesh's very high population and population density. Many projected climate change impacts including sea level rise, higher temperatures (mean temperature increases of 1.4°C and 2.4°C are projected by 2050 and 2100 respectively), evapotranspiration losses, enhanced monsoon precipitation and run-off, potentially reduced dry season precipitation, and increase in cyclone intensity would in fact reinforce many of these baseline stresses that already pose a serious impediment to the economic development of Bangladesh.

Information is drawn from the Country Study (BCAS and DOE, undated), the World Bank study³³ and other sources. Water related impacts of climate change will likely be the most critical for Bangladesh largely related to coastal and riverine flooding, but also enhanced possibility of winter (dry season) drought in certain areas. The effects of increased flooding resulting from climate change will be the greatest problem faced by Bangladesh. Both coastal flooding (from sea and river water), and inland flooding (river/rain water) are expected to increase.

Flooding in Bangladesh is a regular feature and has numerous adverse effects including loss of life through drowning, increased prevalence of disease, and destruction of property. This is because much of Bangladesh is located on a floodplain of three major rivers and their numerous tributaries. Looking into the future, climate change is likely to exacerbate flooding for a number of reasons including the following:

Change in frequency and intensity of cyclones: Bangladesh currently has extreme vulnerability to cyclones, both on account of its somewhat unique location and topography (that creates an inverted funnel effect), and because of the low (though growing) capacity of its society and institutions to cope with such extreme events. Cyclones originate in the deep Indian Ocean and track through the Bay of Bengal where the shallow waters contribute to huge tidal surges when cyclones make landfall. Existing literature records storm surges in the range of 1.5 to 9 meters and some sources even cite particular cyclones as having resulted in surges almost 15 m in height. Given that over two-thirds of the country is less than 5 m above sea-level and densely populated, storm surges contribute to flooding and loss of life and livelihoods far beyond the coast. The intense precipitation that usually accompanies the cyclone only adds to the damage through inland and riverine flooding. A

cyclone in 1970 resulted in close to 300,000 deaths, and another in 1991 led to the loss of 138,000 lives, although in recent years greater success in disaster management has significantly reduced the lives lost³³. Nevertheless, the potential for economic and infrastructural damage remains very significant.

Sea level rise: Another critical variable that determines the vulnerability of Bangladesh to climate change impacts is the magnitude of sea level rise. There is no specific regional scenario for net sea level rise in part because the Ganges-Brahmaputra delta is still active and the morphology is highly dynamic. Literature suggests that the coastal lands are receiving additional sediments due to tidal influence, while there are parts where land is subsiding due to tectonic activities³⁵. Since the landform is constituted by sediment decomposition, compaction of sediment may also play a role in defining net change in sea level along the coastal zone.

Higher mean sea levels are likely to compound the enhanced storm surges expected to result from cyclones with higher intensity. Even in non-cyclone situations, higher mean sea levels are going to increase problems of coastal inundation and salinization in the low-lying deltaic coast. Sea level rise will result in coastal flooding both under ambient conditions (given the low elevations of the coast), and even more so in the event of storm surges. It will also indirectly cause riverine flooding by causing more backing up of the Ganges-Brahmaputra- Meghna rivers along the delta.

Increased precipitation: While this is not certain, the climate models tend to show increased precipitation, particularly during the monsoon season. This will contribute to increased runoff. For example, Mirza and Dixit²¹ found that 2°C warming with a 10% increase in precipitation (close to the mean GCM projection for 2100 June-July-August) would increase runoff in the Ganges, Brahmaputra, and Meghna rivers by 19%, 13% and 11% respectively.

Increased glacier melt: Nepal case study shows higher temperatures will result in more glacial melt, increasing runoff from the neighboring Himalayas into the Ganges and Brahmaputra rivers. Given the altitude of the mountains and the enormous size of the glaciers, this problem will most likely continue over the century. The problem could be of even greater concern as there is evidence to show that temperatures in the Himalayas (where the glaciers are located) are rising at higher rates, thereby contributing to enhanced snow melt.

Climate change impact on Rural and Coastal Infrastructure

Bangladesh's topographic and climatic systems make it one of the most water-related disaster prone countries in the Asian region. Its unique geographical location makes it especially vulnerable to excessive rainfall both in the catchment area and within the country and predisposes to annual flooding which at times becomes catastrophic (as in

1988 and 1998). Floods have caused a greater loss of life and property and have affected the livelihood of more families and communities in Bangladesh than all other natural hazards combined. In a poverty-stricken country like Bangladesh, resources are always less than needed, especially in a disaster situation, as found in Ahmed's research².

About half the population of Bangladesh falls below the poverty line. Under the prevailing socioeconomic circumstances, it is easily understandable that the poor do not have good-quality houses. Moreover, natural disasters often take a huge toll on poorly built houses and sanitation infrastructure as mentioned in Ahmad's study². Human settlements are, therefore, highly vulnerable to climate change-induced floods and cyclonic storm surges.

Floods, especially the high intensity floods, often devastate physical infrastructure such as road networks, educational centers, market places, administrative buildings etc.¹⁶⁻¹⁸ Even flood protection embankments are threatened to be breached during high intensity floods. The high intensity cyclone of 1991 caused large-scale destruction to coastal embankments along the Patina area in Chittagong, destroyed runways of Chittagong airport, ceased port activities by sinking a number of ships in the Patina channel and also in offshore anchorage¹⁹.

The Deluge of 1998 rendered most parts of Dhaka inaccessible by motorized vehicles while the flood waters of 1988 penetrated the runways of Dhaka International Airport and disconnected it for about 11 days from the rest of the world. The telecommunication network was torn off during the cyclone of 1991 and the entire coastal belt was disconnected for weeks.

Climate change induced high intensity events posing huge threats to existing physical infrastructure. Damage to national highways due to flood alone is estimated at 1011 and 3,315 kilometers by the year 2030 and 2050 respectively. The corresponding damage to embankments is estimated at 4,271 and 13,996 kilometers by the year 2030 and 2050 respectively. The aggregated damage figures for health centres and hospitals due to floods, cyclones, sea-level rise and salinity intrusion are estimated at 1,682 and 5,212 respectively for the above two time horizons²⁰.

Bangladesh's coastal resources are ranked as next most vulnerable because the country exists mainly in a delta with most of its population and resources at low elevations and the Sundarbans are threatened by sea level rise. The Sundarbans are important because they are the largest mangrove system in the world and sea level rise could destroy or fundamentally change the entire ecosystem. Sea level is likely to rise; indeed it is more certain than increased flooding. However, the full impacts of sea level rise may not be realized for many decades, thus yielding it second place in the risk ranking.

The major port of Mongla would be at risk, as would one-eighth of the country's agricultural land and 8,000 km of roads²¹. Figures 1 to 8 show the adverse impact of climate change in rural area.

Adaptation Solutions

Adaptation is a process through which societies make themselves better able to cope with an uncertain future. Adapting to climate change entails taking the right measures to reduce the negative effects of climate change (or exploit the positive ones) by making the appropriate adjustments and changes²².



Fig. 1: Rising Water Level: Extra Earth over BWDB Polder Embankment to protect water intrusion at Batiaghata Upazila, Khulna



Fig. 2: Extreme Weather Event: Devastation of Cyclone 'Sidr'

**Fig. 3: Flooding****Fig. 4: Drought****Fig. 5: Salinity Intrusion: Change of Livelihood and land use by Shrimp Culture at Mongla, Bagerhat****Fig. 6: Impact of Salinity: Rapid Deterioration of Reinforced Concrete Structure in Koira Upazila Parishad Bhavan, Khulna****Fig. 7: Salinity Intrusion: Traditional Rainwater Harvesting: Mongla, Khulna****Fig. 8: Unhygienic Toilet in a Rural Market at Mongla Upazila, Bagerhat**

Adaptation options available for management of coastal flooding: Bangladesh is already vulnerable to coastal flooding, and this vulnerability will increase under climate change due to a combination of factors. Bangladesh already employs coastal embankment towards management of coastal flooding, particularly when it is caused by high tides

and storm surges. However, inadequate drainage infrastructure along an embankment can be counter-productive and could interact with several aspects of climate change to produce a cascade of adverse consequences that could in fact enhance the vulnerability of the coastal areas in Bangladesh.

A first order adaptation to climate change would therefore be to build or maintain appropriate drainage infrastructure along coastal embankments. In fact, flow regulators had already been incorporated in the design of existing embankments. However, in many cases the required number of regulators were not built as per design. In other cases, even if the regulators were built, they lacked proper maintenance and consequently failed to serve their intended purpose. The failure of regulators in polder 20 number 24, located in the western coastal region, caused saline flooding for over a decade. It caused severe damage to the agroecology within the embankment and resulted in widespread dislocation of population.

Therefore, building of new drainage regulators along coastal embankments needs to be complemented by an assessment of the need for refurbishing existing regulators followed by their periodic monitoring and maintenance. The participation of local communities would be critical for the effective monitoring and maintenance of coastal embankments and flow regulators. The National Water Policy^{23,24} has given a clear mandate for the formation of associations of water users and water managers, and the participation of these local level organizations at all levels of planning and execution of projects, and more importantly, allowing them to take part in operations and maintenance activities.

While coastal embankments have flow regulators (albeit poorly maintained), the coastal roads network in Bangladesh generally lacks appropriate drainage infrastructure, a factor which is believed to have contributed to the flood of 2000²⁵. Most of the newly built feeder roads along the coastal areas, buildings of which did not usually require rigorous planning and design done with local-level inputs, have completely ignored the necessity of having drainage infrastructure such as culverts, bridges and regulators. Construction of these drainage infrastructure offers a good adaptation option that would certainly reduce flood related vulnerability.

Another family of physical adaptation measures could revolve around enhancing the drainage and/or conveyance capacity of the coastal rivers. This could involve

excavation/dredging of silting rivers to unclog their waterways. Controlled flooding to enhance sedimentation and thereby raise the floodplain further upstream is another adaptation measure that could enhance drainage by increasing the flow gradient. This measure has already been tested under the Khulna-Jessore Drainage Rehabilitation Project²⁶. Raising of the floodplain upstream helped drain the excess water, which in turn reduced flood vulnerability. Post project appraisals have concluded that this 'tidal basin' concept is acceptable to the local population.

Another adaptation measure would involve the use of lifting pumps to take out excessive water from the flood affected areas may be considered as a physical adaptation. Since this involves high costs, it is considered only to save high value properties, infrastructure, urban centers and industrial zones. Pumps can also be used for the purpose of desalination of high value agricultural lands. Repeated flushing of saline affected lands by freshwater and simultaneous disposal of excessive water can reduce soil salinity. Following the high intensity cyclonic event of 1991, Ganoshasthokendra (an NGO) tried such a measure to desalinate a few hectares of land inside the embankment in Maheshkhali Island^{27,28}.

However, the cost of the entire operation was high, thereby reducing its financial viability. The same NGO however also desalinated almost all the salinity affected tube wells after the 1991 cyclone. The operation was quickly completed and allowed people to have fresh potable water. Pumping option as an adaptation may, therefore, be considered to solve certain specific problems (such as salinization of potable water reservoirs) that are expected to occur under climate change.

Figures 9 to 14 illustrate some of the adaptation measures taken by the Government of Bangladesh.

Finally, the ongoing trend towards more effective disaster early warning and response in Bangladesh is also a viable adaptation strategy for flooding that might result from enhanced cyclone intensity that is projected under climate change.



Fig. 9: Protection of Salinity Intrusion and Drainage Improvement



Fig. 10: Cyclone Shelters

**Fig. 11: Killah (Raised Ground) for Livestock Refuse****Fig. 12: Healthy Living Environment****Fig. 13: Public Toilet****Fig. 14: Sanitation Improvement**

Strategies for the development of the Rural Infrastructural System

The strategies for the development of the road system include updating of a road master plan, adoption of rural roads and structures maintenance strategy, maintenance plan and accordingly higher priority to maintenance over new construction, exploring technological options to construct quality roads with available construction materials, introduction of measures to stop overloading, adoption of procedures to maximize generation of employment for the poor, ensuring quality of construction, more involvement of Local Government Institutions (LGI) and ensuring utilization and maintenance of constructed facilities. Improvement of road safety engineering in rural roads leads to minimize road accidents.

Routine periodic emergency maintenance is done by LGED. Vulnerability mapping, climate monitoring, prediction and timely early warning of the extreme climate events are therefore, some of the best strategies for mitigating their negative impact on humanity, property and the environment as well as taking advantage of any positive impacts. Accurate and timely information on the characteristics of the extreme climate events including the associated

vulnerabilities are, therefore, crucial inputs in sustainable development planning.

Climate change is happening and it is caused largely by human activity. Its impacts are beginning to be felt and will worsen in the decades ahead unless we take action. The increasing rate of global warming, courtesy of carbon dioxide and other greenhouse gas emissions from human activities have led to climatic changes and environmental degradation which in turn have resulted in great challenges in relation to diseases and human health. Many diseases which were previously unknown in certain climatic zones are now finding their way to such areas due to changes in the weather conditions. Further, many diseases that had been thought extinct are reemerging in areas with altered climatic conditions that favor their comeback.

It is therefore important that stakeholders and decision makers at industrial, government and international policy levels come up with stringent and workable means of cutting down on greenhouse gases emission to combat the spread of global warming effects, and the resultant climate change which has produced devastating impacts especially among poorer nations. Further, there should be increased funding of

adaptation and coping programs and projects in affected areas to minimize the impacts on human health and curtail the spread of diseases. The purpose of this analysis is to assess the degree of exposure of current development activities and projects to climate risks, as well as the degree of current attention by the government and donors to incorporating such risks in their planning.

A center of excellence named CReLIC (Climate Resilient Local Infrastructure Center) has planned to establish in LGED from Green Climate Fund (GCF). This center will be a knowledge hub regarding climate in Bangladesh. In Bangladesh, there are the different design formats of different departments. This center will research on the all design format and will recommend a unique design format to each sector of climate related issues.

Conclusion

Bangladesh is critically vulnerable to climate induced hazards. Between thirty to seventy per cent of the country is normally flooded each year. The huge sediment loads brought by rivers coupled with a negligible flow gradient add to drainage congestion problems and exacerbate the extent of flooding. Many projected climate change impacts including sea level rise, higher temperatures and evapo-transpiration losses, enhanced monsoon precipitation and run-off, potentially reduced dry season precipitation, and increase in cyclone intensity would in fact reinforce many of these baseline stresses that already pose a serious impediment to the economic development of Bangladesh.

These analyses were conducted in country based on a review of past, ongoing, and planned activities that bear upon the capacity of these two systems to adapt to anticipated impacts of climate change. This was supplemented by interviews by a case study consultant with individuals from key government agencies, NGOs, as well as local stakeholders. The study reviewed donor portfolios and projects, as well as development priorities of the Government of Bangladesh (GOB) to determine the degree of attention to potential risks posed by climate change on relevant sectors.

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